

Conjecture 4.1 *The following decision problem is NP-Complete: “Given an input graph G , determine if $rc(G) = 2$ ”.*

Notice that the problem trivially belongs to NP. More general, and therefore perhaps easier, is the following conjecture:

Conjecture 4.2 *Computing $rc(G)$ is NP-hard.*

- Theorem 1.5 asserts that minimum degree $n/2 + \log n$ guarantees $rc(G) = 2$. Clearly, minimum degree $n/2 - 1$ does not, as there are connected graphs with minimum degree $n/2 - 1$ and diameter 3 (just take two vertex-disjoint cliques of order $n/2$ each and connect them by a single edge. It is therefore interesting to raise:

Problem 4.3 *Determine the minimum degree threshold that guarantees $rc(G) = 2$.*

- As noted in Section 2, finding the optimal bound for $rc(G)$ given that a graph has minimum degree k seems to be an intriguing open problem.

References

- [1] N. Alon and J. H. Spencer, **The Probabilistic Method**, *Second Edition*, Wiley, New York, 2000.
- [2] B. Bollobás, **Extremal Graph Theory**, Academic Press, London, 1978.
- [3] B. Bollobás and A. Thomason, *Threshold functions*, *Combinatorica* 7 (1986), 35–38.
- [4] Y. Caro, D. West, and R. Yuster, *Connected domination and spanning trees with many leaves*, *Siam J. Disc. Math.* 13 (2000), 202–211.
- [5] Y. Caro and R. Yuster, *Dominating a family of graphs with small connected subgraphs*, *Combin. Probab. Comput.* 9 (2000), 309–313.
- [6] G. Chartrand, G. L. Johns, K. A. McKeon, and P. Zhang, *Rainbow connection in graphs*, *Math. Bohem.*, to appear.
- [7] E. Friedgut and G. Kalai, *Every monotone graph property has a sharp threshold*, *Proc. Amer. Math. Soc.* 124 (1996), 2993–3002 .
- [8] J. Petersen, *Die theorie der regularen graphen*, *Acta Math.* 15 (1891), 193–220.